

Question Set 6

Special random variables

1. Earthquakes occur in a given region in accordance with a Poisson process with rate 5 per year.
 - (a) What is the probability there will be at least two earthquakes in the first half of 2010?
 - (b) Assuming that the event in part (a) occurs, what is the probability that there will be no earthquakes during the first 9 months of 2011?
 - (c) Assuming that the event in part (a) occurs, what is the probability that there will be at least four earthquakes over the first 9 months of the year 2010?
2. The lifetimes of interactive computer chips produced by a certain semiconductor manufacturer are normally distributed having mean 4.4×10^6 hours with a standard deviation of 3×10^5 hours. If a mainframe manufacturer requires that at least 90 percent of the chips from a large batch will have lifetimes of at least 4.0×10^6 hours, should he contract with the semiconductor firm?
3. The height of adult women in the United States is normally distributed with mean 64.5 inches and standard deviation 2.4 inches. Find the probability that a randomly chosen woman is
 - (a) less than 63 inches tall;
 - (b) less than 70 inches tall;
 - (c) between 63 and 70 inches tall;
 - (d) Alice is 72 inches tall. What percentage of women is shorter than Alice?
 - (e) Find the probability that the average of the heights of two randomly chosen women exceeds 66 inches.
 - (f) Repeat part (e) for four randomly chosen women.
4. An IQ test produces scores that are normally distributed with mean value 100 and standard deviation 14.2. The top 1 percent of all scores are in what range?
5. The time (in hours) required to repair a machine is an exponentially distributed random variable with parameter $\lambda = 1$.
 - (a) What is the probability that a repair time exceeds 2 hours?
 - (b) What is the conditional probability that a repair takes at least 3 hours, given that its duration exceeds 2 hours?
6. The number of years a radio functions is exponentially distributed with parameter $\lambda = \frac{1}{8}$. If Jones buys a used radio, what is the probability that it will be working after an additional 10 years?

Joint Distributions and Expectation

1. Suppose that the Rockwell hardness X and abrasion loss Y of a specimen have a joint density given by

$$f_{XY}(u, v) = \begin{cases} u + v & \text{for } 0 \leq u, v \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Find the marginal densities of X and Y . (b) Find $E(X)$ and $\text{Var}(X)$.
2. A product is classified according to the number of defects it contains and the factory that produces it. Let X_1 and X_2 be the random variables representing the number of defects per unit (values 0, 1, 2, or 3) and the factory number (values 1 or 2).

$X_1 \setminus X_2$	1	2
0	1/8	1/16
1	1/16	1/16
2	3/16	1/8
3	1/8	1/4

- (a) Find marginal distributions of X_1 and X_2 . (b) Find $E[X_1]$, $E[X_2]$, $\text{Var}(X_1)$, $\text{Var}(X_2)$, $\text{Cov}(X_1, X_2)$.
3. If $E[X] = 2$ and $E[X^2] = 8$, calculate (a) $E[(2 + 4X)^2]$ and (b) $E[X^2 + (X + 1)^2]$.
 4. The joint density of X and Y is given by

$$f(x, y) = \begin{cases} xe^{-(x+y)} & x > 0, y > 0 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Compute the density of X . (b) Compute the density of Y . (c) Are X and Y independent?
5. The joint density of X and Y is

$$f(x, y) = \begin{cases} 2 & 0 < x < y, 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Compute the density of X . (b) Compute the density of Y . (c) Are X and Y independent?
6. The joint probability density function of X and Y is given by

$$f(x, y) = \frac{6}{7} \left(x^2 + \frac{xy}{2} \right), \quad 0 < x < 1, 0 < y < 2$$

- (a) Verify that this is indeed a joint density function. (b) Compute the density function of X . (c) Find $P\{X > Y\}$.
7. Suppose you are expecting a message past 5 P.M. X , the number of hours after 5 P.M. until it arrives, has PDF:

$$f(x) = \begin{cases} \frac{1}{1.5} & \text{if } 0 < x < 1.5 \\ 0 & \text{otherwise} \end{cases}$$

Find the expected time of arrival.

8. The joint density function of X and Y is given by

$$f(x, y) = \begin{cases} 2e^{-x}e^{-2y} & 0 < x < \infty, 0 < y < \infty \\ 0 & \text{otherwise} \end{cases}$$

Compute (a) $P\{X > 1, Y < 1\}$; (b) $P\{X < Y\}$; and (c) $P\{X < a\}$.

9. Suppose that 15 percent of the families in a certain community have no children, 20 percent have 1, 35 percent have 2, and 30 percent have 3 children; suppose further that each child is equally likely (and independently) to be a boy or a girl. If a family is chosen at random from this community, then B , the number of boys, and G , the number of girls, in this family will have the joint probability mass function shown in Table below.

$$P\{B = i, G = j\}$$

$i \setminus j$	0	1	2	3	Row Sum $= P\{B = i\}$
0	.15	.10	.0875	.0375	.3750
1	.10	.175	.1125	0	.3875
2	.0875	.1125	0	0	.2000
3	.0375	0	0	0	.0375
Column Sum =	.3750	.3875	.2000	.0375	
$P\{G = j\}$					

Determine marginal pmfs and correlation coefficient.